Maximum Sum of K Non-Overlapping Fixed-Length Subarrays

HackerRank

You are given an array A of N integers, and three integers K, L, and M.

You must select exactly K pairwise non-overlapping contiguous subarrays from A. Each selected subarray must have a length of exactly L.

For each selected subarray, you must evaluate its score using exactly one of the following two methods:

Standard Sum: The sum of all elements in the subarray. Alternating Sum: The sum of the elements in the subarray with strictly alternating signs, starting with a positive sign. For a subarray starting at index i, this evaluates to A[i] − A[i+1] + A[i+2] − ⋯ + (−1)^(L−1) A[i+L−1].

You are permitted to use the Alternating Sum evaluation method for at most M of your K selected subarrays. The remaining chosen subarrays must use the Standard Sum method.

Find the maximum possible total score achievable by optimally selecting the subarrays and their evaluation methods.

Input Format

The first line contains a integer, N, denoting the number of integers in array A.

The second line contains a integer, K, denoting the exact number of subarrays to select.

The third line contains a integer, L, denoting the exact length of each subarray.

The fourth line contains a integer, M, denoting the maximum number of times the alternating sum can be applied.

The fifth line contains N space-separated integer denoting the array A[i].

Constraints

1 <= N <= 10^4

1 <= K <= 50

1 <= L <= N/K

0 <= M <= K

-10^5 <= A[i] <= 10^5

Output Format

The output contains an integer value

Sample Input 0

```
5
1
3
1
-10 5 -10 5 -10
```

Sample Output 0

```
20
```

Explanation 0

We must pick 1 subarray of length 3. We can use up to 1 alternating sum. If we pick the subarray from index 0 to 2 (−10, 5, −10): The standard sum is −10 + 5 + −10 = −15. The alternating sum is −10 − 5 + −10 = −25. If we pick the subarray from index 1 to 3 (5, −10, 5): The standard sum is 5 + −10 + 5 = 0. The alternating sum is 5 − −10 + 5 = 20. If we pick the subarray from index 2 to 4 (−10, 5, −10): The standard sum is −15, alternating is −25. The maximum score is 20, achieved by picking the middle subarray and using the alternating sum.

Sample Input 1

```
6
2
2
1
20 -5 20 -5 20 -5
```

Sample Output 1

```
40
```

Explanation 1

We must pick 2 non-overlapping subarrays of length 2. We can use at most 1 alternating sum. Every adjacent pair in this array is (20, −5). For any pair (20, −5), the standard sum is 20 + −5 = 15. The alternating sum is 20 − −5 = 25. We pick the subarray at indices 0 to 1 and the subarray at indices 2 to 3. We use the alternating sum on the first one (score 25) and the standard sum on the second one (score 15). We hit our limit of M=1. Total score is 25 + 15 = 40.

Sample Input 2

```
4
2
2
2
1 2 3 4
```

Sample Output 2

```
10
```

Explanation 2

We must pick 2 subarrays of length 2. The only way to fit them is picking indices 0 to 1 and indices 2 to 3. For (1, 2), standard sum is 3, alternating sum is $1 - 2 = -1$. For (3, 4), standard sum is 7, alternating sum is $3 - 4 = -1$. Although we are allowed to use up to 2 alternating sums, we are not forced to. Using the standard sum for both yields the best result. Total score is $3 + 7 = 10$.